7 Vector-Valued Functions

7.1 Coordinate Systems in 3D Space

Specifying a point or giving directions in space can be accomplished in multiple ways:

- Giving a direction and a distance.
- Using Cartesian coordinates $(d_1, d_2, d_3) = (x, y, z)$.
- Specifying an angle relative to a reference axis along with a magnitude.
- Referencing a relative landmark.
- Using geographic coordinates like latitude and longitude.

7.1.1 Polar Coordinates (2D)

In the xy-plane, a point is specified by its radial distance r from the origin and an angle θ relative to the positive x-axis: (r, θ).

$$r = \sqrt{x^2 + y^2} \quad (r \geq 0)$$

While $\theta = \arctan\left(\frac{y}{x}\right)$ is commonly used, it is undefined when $x = 0$. The foundational algebraic transformation between systems is given by:

$$x = r\cos\theta$$

$$y = r\sin\theta$$

7.1.2 Cylindrical Coordinates

Cylindrical coordinates (r, θ, z) extend polar coordinates into 3D space by preserving the standard Cartesian z-axis:

$$x = r\cos\theta$$
$$y = r\sin\theta$$
$$z = z$$

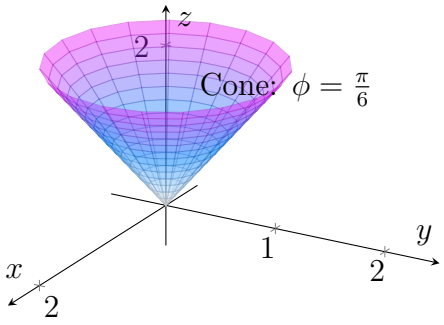
7.1.3 Spherical Coordinates

Spherical coordinates are represented as (ρ, θ, ϕ), where ρ is the distance from the origin, θ is the azimuthal angle in the xy-plane, and ϕ is the polar angle measured down from the positive z-axis.

$$x = \rho\sin\phi\cos\theta$$
$$y = \rho\sin\phi\sin\theta$$
$$z = \rho\cos\phi$$

The standard domain boundaries are:

$$\rho \geq 0$$
$$0 \leq \theta < 2\pi$$
$$0 \leq \phi \leq \pi$$



7.2 Coordinate System Transformations

Problem 7.1: Conversion Between Coordinate Systems

Convert the surface equation $x^2 + \dfrac{y^2}{a^2} = z^2$ into cylindrical and spherical coordinates.

Solution 7.1

Cylindrical Coordinates:

Substituting $x = r\cos\theta$, $y = r\sin\theta$, and $z = z$ into the original equation yields:

$$\begin{aligned}(r\cos\theta)^2 + \frac{(r\sin\theta)^2}{a^2} &= z^2 \\ r^2\cos^2\theta + \frac{r^2\sin^2\theta}{a^2} &= z^2 \\ r^2\left(\cos^2\theta + \frac{\sin^2\theta}{a^2}\right) &= z^2\end{aligned}$$

Spherical Coordinates:

Substituting $x = \rho\sin\phi\cos\theta$, $y = \rho\sin\phi\sin\theta$, and $z = \rho\cos\phi$ into the original equation yields:

$$\begin{aligned}(\rho\sin\phi\cos\theta)^2 + \frac{(\rho\sin\phi\sin\theta)^2}{a^2} &= (\rho\cos\phi)^2 \\ \rho^2\sin^2\phi\cos^2\theta + \frac{\rho^2\sin^2\phi\sin^2\theta}{a^2} &= \rho^2\cos^2\phi \\ \rho^2\sin^2\phi\left(\cos^2\theta + \frac{\sin^2\theta}{a^2}\right) &= \rho^2\cos^2\phi\end{aligned}$$

Assuming $\rho \neq 0$, we divide both sides by ρ^2:

$$\sin^2\phi\left(\cos^2\theta + \frac{\sin^2\theta}{a^2}\right) = \cos^2\phi$$
$$\cos^2\theta + \frac{\sin^2\theta}{a^2} = \frac{\cos^2\phi}{\sin^2\phi}$$
$$\cos^2\theta + \frac{\sin^2\theta}{a^2} = \cot^2\phi$$

7.3 Vector-Valued Functions

Definition 7.1: Vector-Valued Functions

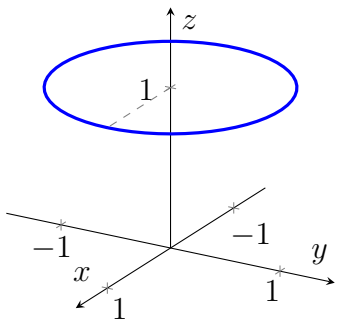
A vector-valued function maps one or more real parameters to a space vector:

$$\vec{r}(t) = \langle x(t), y(t), z(t)\rangle\,, \quad \vec{r} : I \to \mathbb{R}^3$$

where $I \subseteq \mathbb{R}$ is an interval. This traces out a spatial curve parametrized by t.

Example 7.1.

Consider the vector function $\vec{r}(t) = \langle \cos t, \sin t, 1\rangle$ for $0 \leq t < 2\pi$. This describes a unit circle in 3D space elevated to the horizontal plane $z = 1$.



Problem 7.2: Finding Collisions and Intersections

Let $\vec{r}_1(t) = \left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle$ and $\vec{r}_2(t) = \langle t, t^2\rangle$. Find all collision points and structural intersections between the curves.

Solution 7.2

1. Collision Points (Same location at same time t):

Setting $\vec{r}_1(t) = \vec{r}_2(t)$ gives:

$$\left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle = \langle t, t^2 \rangle$$

Equating the individual components gives the system:

$$t^2 + \frac{t}{2} = t$$
$$t - t^3 = t^2$$

Solving the first equation for t:

$$t^2 + \frac{t}{2} - t = 0$$
$$t^2 - \frac{t}{2} = 0$$
$$t\left(t - \frac{1}{2}\right) = 0 \implies t = 0 \quad \text{or} \quad t = \frac{1}{2}$$

Now, we verify both parameters in the second component equation $t - t^3 = t^2$:

$$\text{For } t = 0: \quad 0 - (0)^3 = (0)^2 \implies 0 = 0 \quad \text{(Valid)}$$
$$\text{For } t = \frac{1}{2}: \quad \frac{1}{2} - \left(\frac{1}{2}\right)^3 = \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$$
$$\left(\frac{1}{2}\right)^2 = \frac{1}{4} = \frac{2}{8} \neq \frac{3}{8} \quad \text{(Invalid)}$$

Therefore, $t = 0$ is the only valid collision time. Evaluating $\vec{r}_1(0)$:

$$\vec{r}_1(0) = \left\langle 0^2 + \frac{0}{2}, 0 - 0^3 \right\rangle = \langle 0, 0 \rangle$$

The collision point is $(0, 0)$.

2. Intersection Points (Same location at potentially different times t and

s**):**

Setting $\vec{r}_1(t) = \vec{r}_2(s)$ gives:

$$\left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle = \left\langle s, s^2 \right\rangle$$

Equating components yields:

$$s = t^2 + \frac{t}{2} \tag{1}$$

$$t - t^3 = s^2 \tag{2}$$

Substituting equation (1) into equation (2):

$$\begin{aligned} t - t^3 &= \left(t^2 + \frac{t}{2}\right)^2 \\ t - t^3 &= t^4 + t^3 + \frac{t^2}{4} \\ 0 &= t^4 + 2t^3 + \frac{t^2}{4} - t \\ 0 &= t\left(t^3 + 2t^2 + \frac{t}{4} - 1\right) \end{aligned}$$

Setting each factor to zero:

$$\begin{aligned} t = 0 &\implies s = 0^2 + \frac{0}{2} = 0 \\ &\implies \text{Intersection point } (0, 0) \\ t^3 + 2t^2 + \frac{t}{4} - 1 = 0 &\implies t \approx 0.576 \end{aligned}$$

Solving for s when $t \approx 0.576$:

$$\begin{aligned} s &= (0.576)^2 + \frac{0.576}{2} \\ s &\approx 0.332 + 0.288 \approx 0.620 \end{aligned}$$

Evaluating the spatial coordinates using $\vec{r}_2(s)$:

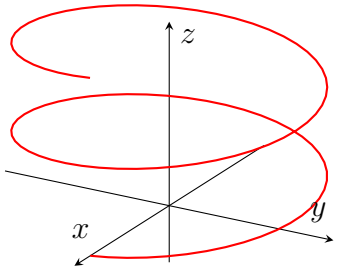
$$x = s \approx 0.620$$

$$y = s^2 \approx (0.620)^2 \approx 0.384$$

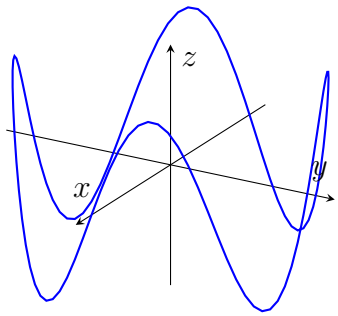
Hence, the two spatial intersection points are $(0, 0)$ and $(0.620, 0.384)$.

7.4 Gallery of Parametric Space Curves

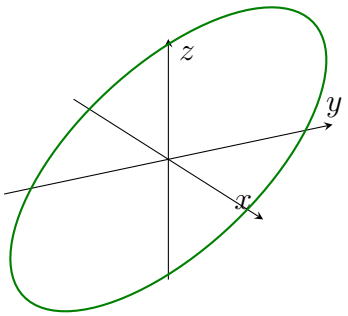
Below are standard parametric 3D space curves used in multivariable trajectory analysis:



$\vec{r}(t) = \langle \cos t, \sin t, t \rangle$



$\vec{r}(s) = \langle \cos s, \sin s, \sin(4s) \rangle$



$\vec{r}(t) = \langle \cos t, \sin t, 4\sin t \rangle$

7.5 Parameterization of Curves and Surfaces

Problem 7.3: Parameterization of a Curve on a Surface

Find a parameterization for the curve given by the intersection of the surfaces $y = \sqrt{x^2 + z^2}$ and $x = z^2$.

Solution 7.3

Notice that for any choice of $z \in \mathbb{R}$, the value of x is uniquely determined by $x = z^2$. We can set $z = t$ as our parameter. Then,

$$x = t^2$$

$$z = t$$

$$y = \sqrt{(t^2)^2 + t^2} = \sqrt{t^4 + t^2} = |t|\sqrt{t^2 + 1}$$

Thus, a vector parameterization of the curve is:

$$\vec{r}(t) = \left\langle t^2, |t|\sqrt{t^2+1}, t \right\rangle, \quad t \in \mathbb{R}$$

7.6 Calculus of Vector-Valued Functions

Calculus operations on vector-valued functions are defined component-wise.

For a vector-valued function $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$, limits, derivatives, and integrals are computed as follows:

$$\lim_{t \to t_0} \vec{r}(t) = \left\langle \lim_{t \to t_0} x(t), \lim_{t \to t_0} y(t), \lim_{t \to t_0} z(t) \right\rangle$$

$$\vec{r}\,'(t) = \langle x'(t), y'(t), z'(t) \rangle$$

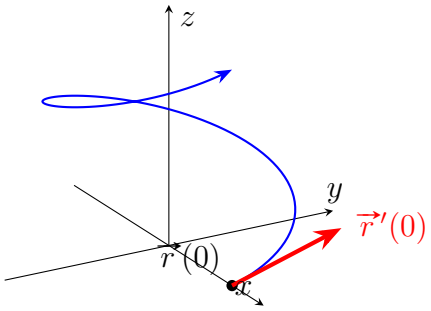
$$\int_a^b \vec{r}(t)\mathrm{d}t = \left\langle \int_a^b x(t)\mathrm{d}t, \int_a^b y(t)\mathrm{d}t, \int_a^b z(t)\mathrm{d}t \right\rangle$$

Example 7.2 (Tangent Vectors and Parameterizing Tangent Lines)**.**
Given the circular helix $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$.

Differentiating component-wise gives:

$$\vec{r}\,'(t) = \langle -\sin t, \cos t, 1 \rangle$$
$$\vec{r}\,'(0) = \langle 0, 1, 1 \rangle$$

The derivative $\vec{r}\,'(0)$ yields a vector representing the tangent direction of the curve at $t = 0$.



To parameterize the tangent line $\ell(t)$ to the helix at $t = 0$:

$$\vec{r}\,(0) = \langle 1, 0, 0 \rangle$$

Thus,

$$\begin{aligned} \ell(t) &= \vec{r}\,(0) + t\,\vec{r}\,'(0) \\ &= \langle 1, 0, 0 \rangle + t\,\langle 0, 1, 1 \rangle = \langle 1, t, t \rangle \end{aligned}$$

Problem 7.4: Derivative and Singular Points of a Cycloid

Find the derivative of the vector-valued function $\vec{r}\,(t) = \langle t - \sin t, 1 - \cos t \rangle$ and evaluate it at $t = \frac{\pi}{2}, \pi, 2\pi$.

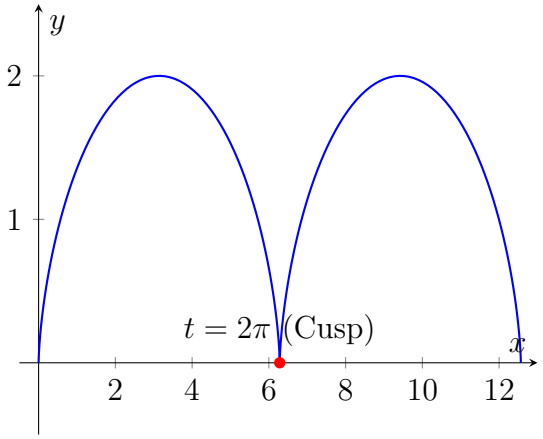
Solution 7.4

Differentiating each component with respect to t:

$$\begin{aligned} \vec{r}\,'(t) &= \left\langle \frac{\mathrm{d}}{\mathrm{d}t}(t - \sin t), \frac{\mathrm{d}}{\mathrm{d}t}(1 - \cos t) \right\rangle \\ &= \langle 1 - \cos t, \sin t \rangle \end{aligned}$$

Evaluating at specific parameter values:

$$\vec{r}'\left(\frac{\pi}{2}\right) = \left\langle 1 - \cos\frac{\pi}{2}, \sin\frac{\pi}{2} \right\rangle = \langle 1, 1 \rangle$$
$$\vec{r}'(\pi) = \langle 1 - \cos\pi, \sin\pi \rangle = \langle 2, 0 \rangle$$
$$\vec{r}'(2\pi) = \langle 1 - \cos 2\pi, \sin 2\pi \rangle = \langle 0, 0 \rangle$$



Notice from the graph of the cycloid that at $t = 2\pi$, the curve forms a sharp sharp corner (cusp). At this point, $\vec{r}'(2\pi) = \langle 0, 0 \rangle$. Although the vector function $\vec{r}(t)$ is differentiable everywhere, the tangent line direction is undefined at the cusp because the derivative vector vanishes.

7.7 Properties of Vector Derivatives & Integration

Problem 7.5: Orthogonality Property for Constant-Length Vector Functions

If $\|\vec{r}(t)\| = c$ for some constant c, what does this imply about the relationship between $\vec{r}(t)$ and its derivative $\vec{r}'(t)$?

Solution 7.5

Since $\|\vec{r}(t)\| = c$, taking the square of both sides yields:

$$\|\vec{r}(t)\|^2 = c^2 \implies \vec{r}(t) \cdot \vec{r}(t) = c^2$$

Differentiating both sides with respect to t using the product rule for dot products:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\vec{r}(t)\cdot\vec{r}(t)\right)=\frac{\mathrm{d}}{\mathrm{d}t}(c^2)$$
$$\vec{r}\,'(t)\cdot\vec{r}(t)+\vec{r}(t)\cdot\vec{r}\,'(t)=0$$
$$2\vec{r}(t)\cdot\vec{r}\,'(t)=0 \implies \vec{r}(t)\cdot\vec{r}\,'(t)=0$$

Therefore, $\vec{r}(t)\perp\vec{r}\,'(t)$ for all t. (The position vector is always orthogonal to its velocity vector).

Product Rule Verification for Dot Products

Given two vector-valued functions $\vec{r_1}(t),\vec{r_2}(t)\in\mathbb{R}^2$:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left[\vec{r_1}(t)\cdot\vec{r_2}(t)\right]&=\frac{\mathrm{d}}{\mathrm{d}t}\left[x_1(t)x_2(t)+y_1(t)y_2(t)\right]\\&=x_1'(t)x_2(t)+x_1(t)x_2'(t)+y_1'(t)y_2(t)+y_1(t)y_2'(t)\\&=\vec{r_1}\,'(t)\cdot\vec{r_2}(t)+\vec{r_1}(t)\cdot\vec{r_2}\,'(t)\end{aligned}$$

Problem 7.6: Initial Value Problem for Motion

A particle moves with velocity $\vec{v}(t)=\left\langle 3t^2-2,\cos(\pi t),\sqrt{t}\right\rangle$. Find its position vector $\vec{r}(t)$, given the initial condition $\vec{r}(0)=\langle 2,1,1\rangle$.

Solution 7.6

Integrating $\vec{v}(t)$ component-wise:

$$\begin{aligned}\vec{r}(t)&=\int\vec{v}(t)\,\mathrm{d}t\\&=\left\langle\int(3t^2-2)\,\mathrm{d}t,\int\cos(\pi t)\,\mathrm{d}t,\int t^{1/2}\,\mathrm{d}t\right\rangle\\&=\left\langle t^3-2t+C_1,\frac{1}{\pi}\sin(\pi t)+C_2,\frac{2}{3}t^{3/2}+C_3\right\rangle\end{aligned}$$

Applying the initial condition $\vec{r}(0) = \langle 2, 1, 1\rangle$:

$$\vec{r}(0) = \langle C_1, C_2, C_3\rangle = \langle 2, 1, 1\rangle \implies C_1 = 2,\, C_2 = 1,\, C_3 = 1$$

Therefore,

$$\vec{r}(t) = \left\langle t^3 - 2t + 2, \frac{1}{\pi}\sin(\pi t) + 1, \frac{2}{3}t^{3/2} + 1 \right\rangle$$

Note on Circular Motion: If given acceleration $\vec{r}''(t) = \langle -\cos t, -\sin t, 0\rangle$, to uniquely guarantee that $\vec{r}(t)$ describes a circle centered at the origin, appropriate initial conditions for both position $\vec{r}(0)$ and velocity $\vec{r}'(0)$ must be specified.

7.8 Arc Length of Curves

The arc length s of a smooth curve $\vec{r}(t) = \langle x(t), y(t)\rangle$ on $t \in [a, b]$ is derived as the limit of polygonal approximations:

$$s = \lim_{\Delta t \to 0} \sum \sqrt{\left(\frac{\Delta x}{\Delta t}\right)^2 + \left(\frac{\Delta y}{\Delta t}\right)^2}\,\Delta t = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2}\,\mathrm{d}t = \int_a^b \|\vec{r}'(t)\|\mathrm{d}t$$

Example 7.3 (Arc Length of a Helix)**.**
Find the arc length of $\vec{r}(t) = \langle \cos t, \sin t, t\rangle$ for $t \in [0, 2\pi]$.

First, evaluate the speed $\|\vec{r}'(t)\|$:

$$\vec{r}'(t) = \langle -\sin t, \cos t, 1\rangle \implies \|\vec{r}'(t)\| = \sqrt{(-\sin t)^2 + \cos^2 t + 1^2} = \sqrt{2}$$

Thus, the arc length is:

$$s = \int_0^{2\pi} \sqrt{2}\,\mathrm{d}t = 2\pi\sqrt{2}$$

Example 7.4 (Arc Length of an Ellipse and Elliptic Integrals)**.**
Consider the ellipse $x^2 + \frac{y^2}{4} = 1$.

Method 1 (Cartesian Parameterization - Upper Half): Setting $x = t$ and $y =$

$2\sqrt{1-t^2}$ for $t \in [-1, 1]$ yields:

$$\vec{r}(t) = \left\langle t, 2\sqrt{1-t^2} \right\rangle \implies \vec{r}\,'(t) = \left\langle 1, -\frac{2t}{\sqrt{1-t^2}} \right\rangle$$

$$\|\vec{r}\,'(t)\| = \sqrt{1 + \frac{4t^2}{1-t^2}} = \sqrt{\frac{1+3t^2}{1-t^2}}$$

The arc length of the upper half is given by

$$\int_{-1}^{1} \sqrt{\frac{1+3t^2}{1-t^2}}\, \mathrm{d}t.$$

Method 2 (Trigonometric Parameterization - Full Ellipse): Setting $x = \cos t$ and $y = 2\sin t$ for $t \in [0, 2\pi]$:

$$\vec{r}\,'(t) = \langle -\sin t, 2\cos t \rangle \implies \|\vec{r}\,'(t)\| = \sqrt{\sin^2 t + 4\cos^2 t} = \sqrt{4 - 3\sin^2 t}$$

The total perimeter of the ellipse is:

$$s = \int_0^{2\pi} \sqrt{4 - 3\sin^2 t}\, \mathrm{d}t$$

Definition 7.2: Elliptic Integral

An elliptic integral is a type of integral that cannot be expressed in terms of elementary functions.

Elliptic Integrals: It turns out these integrals cannot be evaluated in terms of elementary functions. They belong to a class of functions known as **elliptic integrals**, which require numerical methods to evaluate.